Graph Neural Networks for Hardware Reverse Engineering

Interdisciplinary Project in Informatics

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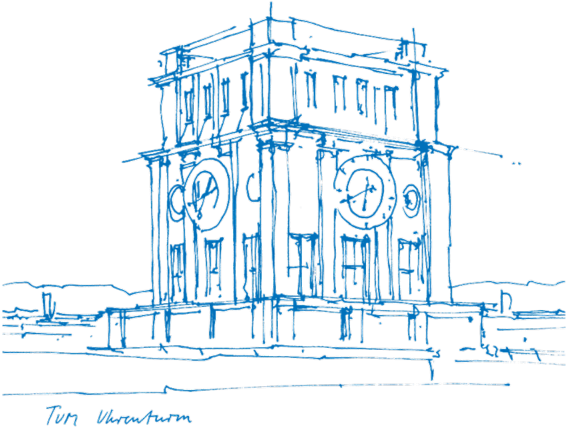
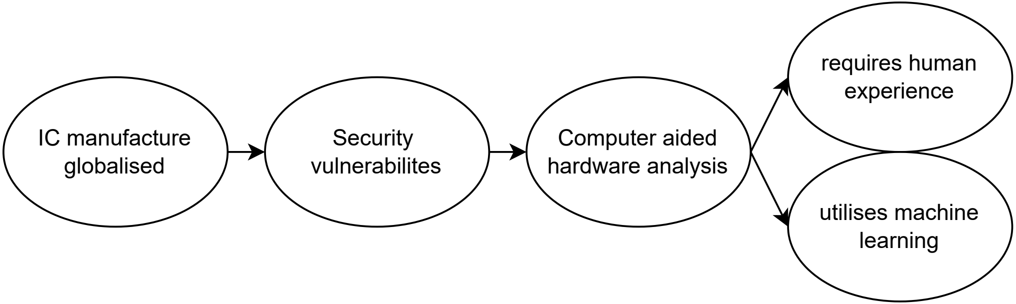


Outline

Introduction



Outline

Research Question

Can we implement the GNN as described in GNN-RE[1] and replicate their results and maintain a result like theirs on larger datasets?

Outline

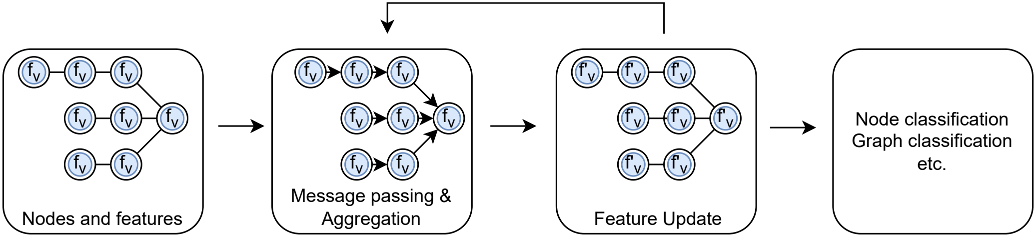
Current Works

Paper	CNN[1]	GNN	
		GCN[2]	GAT[3]
Silva et al. (2018) [7]	x		
Fayyazi et al. (2019) [2]	x		
Kunal et al. (2020) [4]		x	
Hong et al. (2021) [3]		x	
Li et al. (2020) [5]			x

[1] Convolutional Neural Network
[2] Graph Convolutional Neural Network
[3] Graph Attention Network

What is a GNN and how does it work?

- Graph Neural Network first introduced by Scarselli et al. [6]



"GNN-RE: Graph Neural Networks for Gate-Level Netlist Reverse Engineering" by Alrahis et al. (2022) [1]

- GNN-RE: machine learning-based model for reverse engineering
- flatted gate-level netlists are analyzed
- average accuracy of around 98%

GNN-RE - Configuration

Architecture		Training and Sampling	
Total # Layers	5	**Optimizer**	Adam
GNN Depth	1	**Dropout**	0.1
Input Layer	[33, 256]	**Attention**	8
Hidden Layers 1-4	[512, 256]	**Learning Rate**	0.01
Output Layer	[256, *#classes*]	**Sampler**	Random Walk
Aggregation	Attention aggregator with multi-head concatenation	**Walk Length**	2
Activation	*ReLU*	**Root Nodes**	3000
Classification	*softmax*	**Max # Epochs**	2000

Table 1 GNN-RE configuration details, table taken from [1]

GNN-RE - Dataset Overview

Case Study	Datasets	#Classes	#Features	#Nodes	#Circuits	#Training Nodes	#Validation Nodes	#Testing Nodes
I	Single-Modules	4	33	32,528	27	6,318	357	25,853
	Sequential-Modules	3	39	841	29	700	69	72
II	Add-Mul-Mux	3	33	15,582	7	14,467	239	876
	Add-Mul-Mix	2		21,602	6	19,377	576	1,649
	Add-Mul-Combine	2		14,288	6	13,256	217	815
	Add-Mul-All	3		51,472	19	47,100	1,032	3,340
	Add-Mul-Comp	4		15,898	6	14,717	255	926
	Add-Mul-Sub	4		18,206	6	16,786	309	1,111
	Add-Mul-Comp-Sub	5		19,151	6	17,648	342	1,161
	Interconnected-Modules	5		104,727	37	96,251	1,938	6,538

Figure 1 Dataset overview used by the authors, figure taken from [1]

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Figure 2 Dataset overview used by the authors, figure taken from [1]

GNN-RE - Dataset Overview

Case Study	Datasets	#Classes	#Features	#Nodes	#Circuits	#Training Nodes	#Validation Nodes	#Testing Nodes
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Figure 3 Dataset overview used by the authors, figure taken from [1]

GNN-RE - Node Classification Result

Case Study	Dataset	GNN Accuracy (%)	Precision (%)					Recall (%)					F1-Score (%)					Training Time (s)
			ADD	MUL	CTRL	COMP	SUB	ADD	MUL	CTRL	COMP	SUB	ADD	MUL	CTRL	COMP	SUB	
I	Single-Modules	99.28	95.15	99.99	-	98.35	87.29	100	99.34	-	98.73	97.75	97.52	99.67	-	92.66	98.05	1,282.05
II	Add-Mul-Mux	99.32	97.43	99.60	97.50	-	-	96.20	99.60	100	-	-	96.82	99.60	98.77	-	-	2,289.13
	Add-Mul-Mix	99.45	99.37	99.47	-	-	-	98.43	99.85	-	-	-	98.9	99.66	-	-	-	3,119.42
	Add-Mul-Combine	99.02	90.12	100	-	-	-	100	98.92	-	-	-	94.87	99.45	-	-	-	2,297.50
	Add-Mul-All	97.75	88.61	99.5	95	-	-	97.25	98.23	88.37	-	-	92.73	98.72	92.68	-	-	6,838.18
	Add-Mul-Sub	98.02	86.42	99.61	90.91	-	100	93.33	99.74	98.77	-	92.55	89.74	99.67	94.67	-	96.13	2,296.61
	Add-Mul-Comp	99.35	94.59	99.87	97.06	100	-	100	100	97.06	89.13	-	97.22	99.94	97.06	94.25	-	2,275.00
	Add-Mul-Comp-Sub	98.28	100	97.42	100	100	100	100	100	91.14	92.9	100	100	98.69	95.36	96.32	100	2,308.70
	Interconnected-Modules	98.87	97.51	99.14	94.74	100	100	96.24	99.61	97.5	97.39	94.88	96.87	99.38	96.1	98.68	97.37	16,317.44

Figure 4 GNN-RE node classification result, figure taken from [1]

GNN-RE - Node Classification Result

Case Study	Dataset	GNN Accuracy (%)	Precision (%)					Recall (%)					F1-Score (%)					Training Time (s)
			ADD	MUL	CTRL	COMP	SUB	ADD	MUL	CTRL	COMP	SUB	ADD	MUL	CTRL	COMP	SUB	
I	Single-Modules	99.28	95.15	99.99	-	98.35	87.29	100	99.34	-	98.73	97.75	97.52	99.67	-	92.66	98.05	1,282.05
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	Add-Mul-Combine	99.02	90.12	100	-	-	-	100	98.92	-	-	-	94.87	99.45	-	-	-	2,297.50
	Add-Mul-All	97.75	88.61	99.5	95	-	-	97.25	98.23	88.37	-	-	92.73	98.72	92.68	-	-	6,838.18
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	Add-Mul-Comp-Sub	98.28	100	97.42	100	100	100	100	100	91.14	92.9	100	100	98.69	95.36	96.32	100	2,308.70
	Interconnected-Modules	98.87	97.51	99.14	94.74	100	100	96.24	99.61	97.5	97.39	94.88	96.87	99.38	96.1	98.68	97.37	16,317.44

Figure 5 GNN-RE node classification result, figure taken from [1]

GNN-RE - Node Classification Result

Case Study	Dataset	GNN Accuracy (%)	Precision (%)					Recall (%)					F1-Score (%)					Training Time (s)
			ADD	MUL	CTRL	COMP	SUB	ADD	MUL	CTRL	COMP	SUB	ADD	MUL	CTRL	COMP	SUB	
I	Single-Modules	99.28	95.15	99.99	-	98.35	87.29	100	99.34	-	98.73	97.75	97.52	99.67	-	92.66	98.05	1,282.05
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	Add-Mul-Combine	99.02	90.12	100	-	-	-	100	98.92	-	-	-	94.87	99.45	-	-	-	2,297.50
	Add-Mul-All	97.75	88.61	99.5	95	-	-	97.25	98.23	88.37	-	-	92.73	98.72	92.68	-	-	6,838.18
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Figure 6 GNN-RE node classification result, figure taken from [1]

GNN-RE - Node Classification Result

Case Study	Dataset	GNN Accuracy (%)	Precision (%)					Recall (%)					F1-Score (%)					Training Time (s)
			ADD	MUL	CTRL	COMP	SUB	ADD	MUL	CTRL	COMP	SUB	ADD	MUL	CTRL	COMP	SUB	
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Figure 7 GNN-RE node classification result, figure taken from [1]

Outline

Preprocessing

Connected ports		Gate types												Connected wires	
Primary Input	Primary Output	AND	AOI	BUF	DFF	INV	MUX	NAND	NOR	OAI	OR	XNOR	XOR	Incoming wires	Outgoing wires

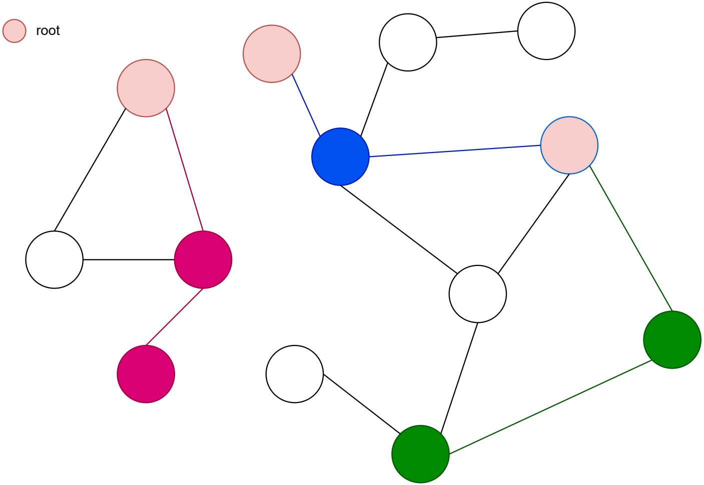
Figure 8 Full feature vector for AISEC data

GNN architecture and training

Architecture		Training and Sampling	
GNN model	GraphSAGE / GAT	**Optimizer**	Adam
Total # Layers	3	**Loss function**	CrossEntropy
Input Layer	[16, 256]	**Attention**	8
Hidden Layers	[256, 256] / [128, 128]	**Learning Rate**	0.01
Output Layer	[256, #num classes]	**Sampler**	Random Walk
Aggregation	Attention aggregator with multi-head concatenation	**Walk Length**	2
Activation	*ReLU*	**Root Nodes**	3000
Classification	*softmax*	**Max # Epochs**	2000 / 500
Dropout Rate	0.1	**Weight decay**	1e-5

Table 2 GNN architecture and training configuration details

Random walk sampling



Post-processing

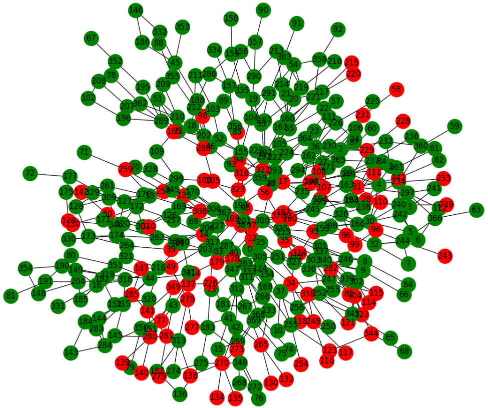


Figure 9 Label correctness before post-processing

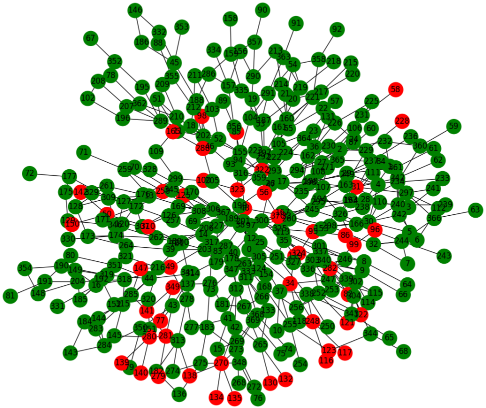


Figure 10 Label correctness after post-processing

Post-processing

Majority based post-processing

vs

Probability based post-processing

Datasets

Dataset Version	#Classes	#Features	#Nodes	#Circuits
as used in GNN-RE [1]	5	33	104,727	37
using our implementation	5	**20**	104,727	37

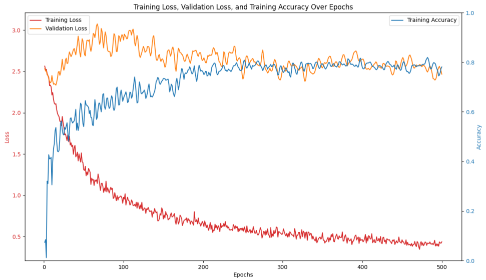
Table 3 GNN-RE dataset [1] differences from feature implementation

Dataset	#Classes	#Features	#Nodes	#Circuits
full dataset	27	16	7,503,589	1,768
- reduced set	13	16	5,548,470	1,247
- arithmetic circuits set	8	16	2,814,577	653
- core circuits set	5	16	2,733,893	594
- osu035 set	13	16	3,328,129	628

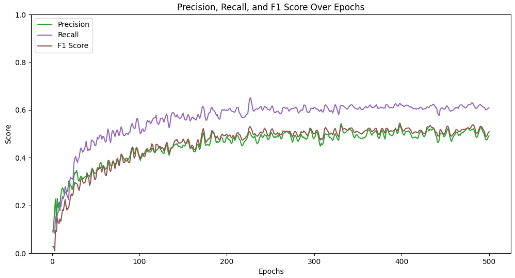
Table 4 AISEC dataset and subsets

Outline

Example test result of the reduced AISEC dataset



(a) Training Accuracy, Validation loss & Training Loss over epochs



(b) Precision, Recall & F1 score over epochs

Figure 11 Metrics evolution during training of GAT on the reduced AISEC dataset

Example subtractor graph from the reduced AISEC dataset with model predicitons

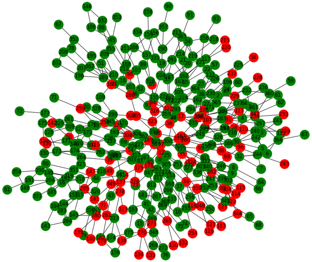


Figure 12 Label correctness before post-processing

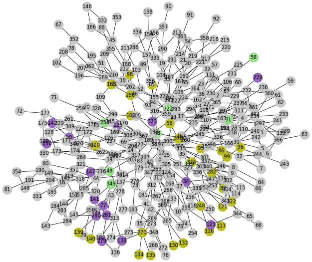


Figure 13 Labeled graph after postprocessing with labels: "sub": silver, "add": purple, "crc": light green, "div": gold

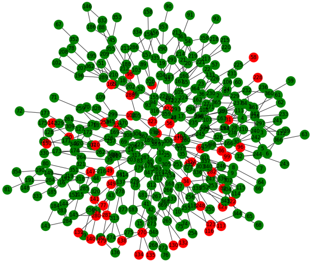


Figure 14 Label correctness after post-processing

GNN performance results of AISEC dataset and subsets

Dataset	GraphSAGE				GAT			
	Accuracy (%)	F1-Score (%)	Precision (%)	Recall (%)	Accuracy (%)	F1-Score (%)	Precision (%)	Recall (%)
GNN-RE	97	93	96	90	97	92	95	90
full dataset	41	20	24	32	34	24	28	44
reduced set	80	58	56	71	72	48	46	59
arithmetic set	59	53	50	68	77	62	57	78
components set	81	62	57	78	95	80	77	85
osu035 set	35	29	37	37	27	21	26	34

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Outline

Discussion

- Importance of high-quality data
- How realistic is perfect data?
- Scalability of random walk sampling

Discussion

- Importance of high-quality data
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Thank you for listening

Any Questions?

Outline

Sources I

[1] Lilas Alrahis et al. "GNN-RE: Graph Neural Networks for Reverse Engineering of Gate-Level Netlists". In: *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems* 41.8 (2022), pp. 2435–2448. DOI: 10.1109/TCAD.2021.3110807.

[2] Arash Fayyazi et al. "Deep Learning-Based Circuit Recognition Using Sparse Mapping and Level-Dependent Decaying Sum Circuit Representations". In: *2019 Design, Automation & Test in Europe Conference & Exhibition (DATE)*. 2019, pp. 638–641. DOI: 10.23919/DATE.2019.8715251.

[3] Xuenong Hong et al. "ASIC Circuit Netlist Recognition Using Graph Neural Network". In: *2021 IEEE International Symposium on the Physical and Failure Analysis of Integrated Circuits (IPFA)*. 2021, pp. 1–5. DOI: 10.1109/IPFA53173.2021.9617311.

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